

DES Pune University, Pune
School of Engineering and Technology
Department of Computer Engineering and Technology
Program: B. Tech in Computer Science and Engineering

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Subject: Python Programming

Assignment No.: 11

Date: 12/01/2026

Lab Assignment: 11

Aim:

Write a program using loop to read the numbers until -1 is encountered. also count the number of prime numbers and composite numbers entered by the user.

Objective:

- To use a while loop for continuous input.
- To identify prime and composite numbers.
- To apply looping and conditional statements in Python.
- To terminate the loop using a sentinel value (-1).

Theory:

A loop in Python is used to execute a block of code repeatedly. The while loop runs as long as the given condition is true.

A prime number is a number greater than 1 having exactly two factors: 1 and itself.

A composite number is a number greater than 1 having more than two factors.

In this program, the user enters numbers continuously until -1 is entered. Each number is checked for prime or composite and the respective count is increased.

Algorithm:

1. Start
2. Initialize prime_count = 0 and composite_count = 0
3. Read a number from the user
4. If the number is -1, stop the loop
5. If the number is less than or equal to 1, skip it
6. Count the number of factors of the number
7. If the number has exactly two factors, increment prime_count
8. Otherwise, increment composite_count
9. Repeat steps 3 to 8
10. Display the prime and composite counts
11. Stop

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Pseudo Code:

```
START
prime_count = 0
composite_count = 0
WHILE True
    READ n
    IF n == -1
        BREAK
    IF n <= 1
        CONTINUE
    count = 0
    FOR i = 1 TO n
        IF n % i == 0
            count = count + 1
    END FOR
    IF count == 2
        prime_count = prime_count + 1
    ELSE
        composite_count = composite_count + 1
END WHILE
PRINT prime_count, composite_count
STOP
```

Code:

```
prime_count = 0;
composite_count = 0;
while True:
    n = int(input("Enter a number (-1 to stop): "));
    if n == -1:
        break;
    if n <= 1:
        continue;
    count = 0;
    for i in range(1, n + 1):
        if n % i == 0:
            count += 1;
    if count == 2:
        prime_count += 1;
    else:
        composite_count += 1;
print("Number of prime numbers:", prime_count)
print("Number of composite numbers:", composite_count)
```

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Output:

```
Enter a number (-1 to stop): 2
Enter a number (-1 to stop): 4
Enter a number (-1 to stop): 11
Enter a number (-1 to stop): 1
Enter a number (-1 to stop): 2
Enter a number (-1 to stop): -1
Number of prime numbers: 3
Number of composite numbers: 1
```

Conclusion:

The program successfully reads numbers until -1 is encountered and correctly counts the number of prime and composite numbers using loops and conditional statements.